

PHASE

REQUIREMENT ANALYSIS PHASE

Date	28 September 2026
Team ID	SWTID-2026-5127
Project name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum marks	4

2. Requirement Analysis Phase

2.1 Purpose of this phase

Requirement analysis converts the approved idea into precise, verifiable statements of what the system must do and how well it must do it. Clear requirements prevent rework, guide design and become the checklist against which testing is performed.

2.2 Requirement gathering approach

Requirements were gathered through:

- **Interviews** with IT helpdesk agents, procurement staff and line managers.
- **Process walkthroughs** of recent laptop orders to identify real delays.
- **Review of past tickets** to see which information was frequently missing.
- **Workshop** to confirm approval rules, standard models and notification needs.

2.3 Scope

In scope

- A Service Catalog item for standard laptop requests.
- Variables to capture request details.
- A Flow Designer flow for approval, procurement task creation, status update and notifications.
- Handling of approval rejection and basic exceptions.
- Portal visibility of request status.
- Basic reporting on request volume and cycle time.

Out of scope

- Custom or high-end workstation orders.
- Direct integration with vendor or ERP systems.
- Asset lifecycle management beyond linking the request.
- Payment processing.

2.4 User roles

Role	Responsibility in the process
Requester	Submits the request and tracks it
Manager (approver)	Approves or rejects the request
Procurement team	Places the order and confirms delivery
IT fulfilment team	Configures and hands over the laptop
Catalog / Flow administrator	Maintains the catalog item and flow

2.5 Functional requirements

ID	Requirement	Priority
FR-01	The system shall provide a catalog item named "Standard Laptop Request" visible in the Service Portal.	High
FR-02	The form shall capture: requested for, laptop model (from approved list), justification, delivery location, needed-by date and cost center.	High
FR-03	Mandatory fields shall be enforced before submission.	High
FR-04	On submission, a Requested Item (RITM) shall be created automatically.	High
FR-05	The flow shall trigger automatically when the request is submitted.	High
FR-06	The flow shall send an approval request to the manager of the "requested for" user.	High
FR-07	If approved, the flow shall create a procurement task assigned to the Procurement group.	High
FR-08	If rejected, the flow shall update the RITM, notify the requester with the reason and close the request.	High
FR-09	The flow shall send notifications on submission, approval, rejection and completion.	High
FR-10	When the procurement task closes, the flow shall create/complete the fulfilment task for IT setup.	Medium
FR-11	The RITM state and comments shall update at each stage.	Medium
FR-12	If the manager does not respond within a defined period, a reminder shall be sent.	Medium
FR-13	A dashboard/report shall show volume, approval time and fulfilment time.	Low

2.6 Non-functional requirements

- **Usability:** the form must be simple enough to complete in under two minutes.
- **Performance:** the flow must start within seconds of submission.
- **Reliability:** failed flow executions must be visible in the execution log for re-run or investigation.
- **Security:** only authorized users may approve; roles restrict who can edit the flow and catalog item.
- **Auditability:** approvals, comments and state changes must be retained.
- **Maintainability:** the flow should use readable names, subflows for reusable logic and minimal scripting.
- **Scalability:** the pattern should be reusable for other hardware requests.

2.7 Business rules and process rules

1. Only the standard models listed in the catalog may be requested.
2. The approver is the direct manager of the person receiving the laptop.
3. If the manager field is empty, the request is routed to a fallback approver (for example, the department head or IT manager).
4. Work does not begin on procurement until approval is granted.
5. Rejected requests require a reason.

2.8 Data requirements

Data element	Source	Notes
Requested for	User table	Reference field
Manager	User record	Used for approval lookup
Laptop model	Catalog variable (select box or reference to a model list)	Limited to standard models
Justification	Catalog variable	Mandatory, multi-line text
Delivery location	Location table	Reference field
Cost center	Catalog variable / user record	Used for budget tracking
Approval outcome and time	Approval record	Audit trail

2.9 Assumptions and dependencies

- The ServiceNow instance is available, with Flow Designer and Service Catalog enabled.
- User, group and location data is accurate.
- The Procurement and IT Fulfilment groups exist and have members.
- Email notification is configured on the instance.

2.10 Constraints

- Development is performed on a non-production or developer instance.
- Time allocated to the project is limited, so scope is restricted to standard laptops.
- The team should prefer out-of-the-box capability over custom code.

2.11 Use case: Request a standard laptop

Item	Description
Actor	Requester
Precondition	User is logged in to the Service Portal
Main flow	1) Open catalog. 2) Select "Standard Laptop Request". 3) Fill in details. 4) Submit. 5) Receive confirmation and request number.
Alternate flow	Missing mandatory field: form displays an error and prevents submission.
Postcondition	RITM created and flow started.

2.12 Acceptance criteria

- A request submitted with valid data creates a RITM and starts the flow.
- Manager receives an approval request; the outcome is recorded.
- Approval creates a procurement task; rejection closes the request with notification.
- Requester receives notifications at defined milestones.
- No manual action is needed between submission and procurement task creation.

2.13 Traceability overview

Each functional requirement is linked to a design element and later to a test case (for example, FR-06 traces to the "Ask for Approval" action and to test case TC-03). This traceability ensures nothing is designed or tested that does not serve a requirement, and no requirement is left untested.

2.14 Outcome of the phase

A signed-off requirement set covering functional needs, non-functional expectations, rules, data, scope and acceptance criteria. The team can now start designing the solution.